# More Multilingual NLP with Interlinear Gloss Corpora

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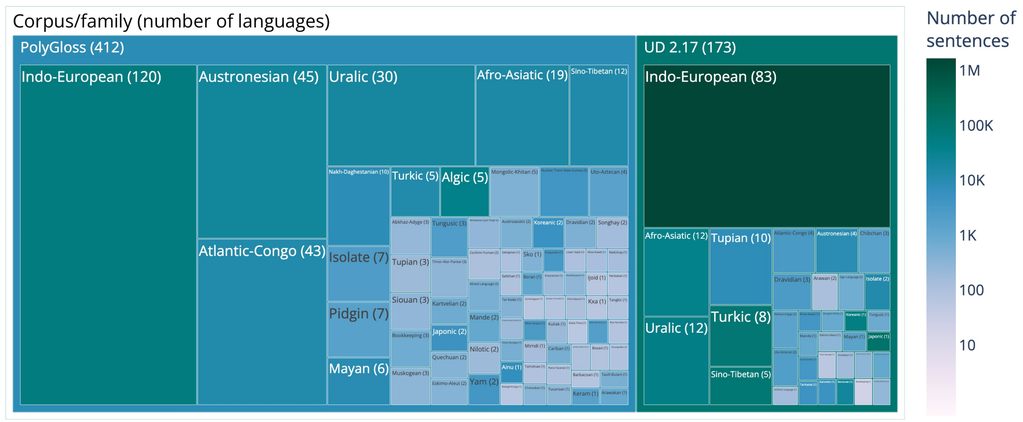
*nabə-ʔi* *bazoʔ* *n'ergö-laʔbə-ʔjə*
duck-PL again fly-DUR-3PL
'The ducks are flying again.'

Kamas [kama1378]. From PolyGloss, wav2gloss subcorpus (*tePKZ_196X_Ducks_flk.071*)

## How can glossed data be used in NLP?

### The PolyGloss corpus [2, 3]

- Compiles IGT examples from grammars, articles and databases
- Varying glossing conventions and analysis levels
- Covers 2067 languages across 177 top-level families
- More than 50 examples for 412 langs across 76 families




**Above:** Comparison of dataset size and genealogical diversity in the Polygloss corpus and Universal Dependencies. Rectangle size represents the number of languages per family; color represents the total number of examples across all languages in a family.

## Automatic gloss prediction

*nabəʔi* *bazoʔ* *n'ergölaʔbəʔjə*
'The ducks are flying again.'

*nabə* (duck) – *ʔi* (PL)
*bazoʔ* (again)
*n'ergö* (fly) – *laʔbə* (DUR) – *ʔjə* (3PL)

### The PolyGloss model [2]

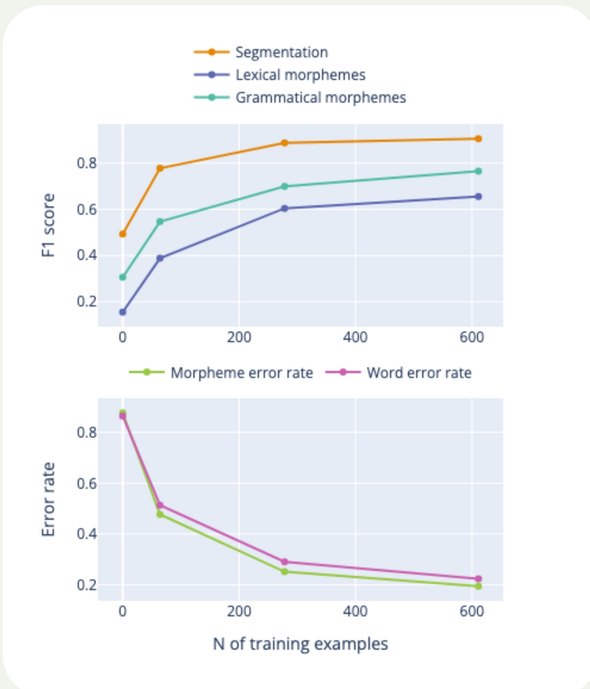
- Multilingual seq2seq model: ByT5 + pretraining on the PolyGloss corpus
- Joint prediction of segmentation and glosses; interleaved format for segment-gloss alignment
- Cross-lingual transfer: multilingual pretraining improves accuracy on languages with less training data (up to 3k examples)

> Predict the glosses and morphological segmentation (in parentheses) for the following text in Kamas.
> Text in Kamas: nabəʔi bazoʔ n'ergölaʔbəʔjə
> Translation into English: 'The ducks are flying again.'
> Gloss:
> *duck(nabə)–PL(ʔi) again(bazoʔ) fly(n'ergö)–DUR(laʔbə)–3PL(ʔjə)*

### Evaluation on Gawarbati

- Indo-Aryan, not in PolyGloss
- Train LoRA adapters (rank 8, 25 epochs) on varying amounts of data

`train`: 1/2/4 texts* (64/278/611 examples)
`test`: 2 texts (194 examples)




**Above:** Performance on Gawarbati test set, by number of examples used in LoRA adapter training.

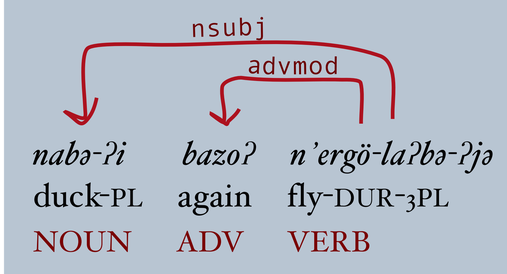
### Why?

- 'Rough gloss' longer texts -> human review/post-editing
- Stemming and morphological analysis as part of NLP pipeline for more languages

### Next steps

- Compare to baselines: hard-attention neural model [4] and base ByT5 model
- Qualitative error analysis: which morphemes are hardest?

## Inferring syntactic annotations from glosses

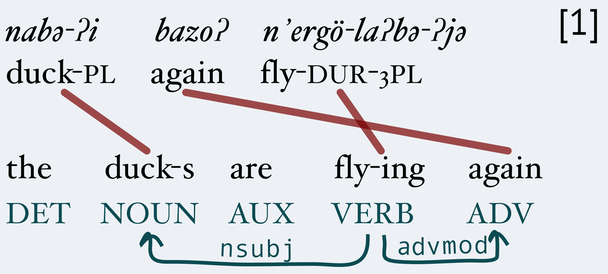



### Strategies for inference

**Feature heuristics** [1]

```
stem = again  -> ADV
gram = DUR-3PL-> VERB
gram = NOM    -> nsubj
```

**Word alignment -> annotation transfer** [1]




**LLM prompting**

> Given the following interlinear glossed example in Kamas, identify the parts of speech for all words. Then, identify the positions of the subject (nsubj) and its head, the verbal predicate (root), in the main clause.
> Text in Kamas: nabə-ʔi bazoʔ n'ergö-laʔbə-ʔjə
> Gloss: duck–PL again fly–DUR–3PL
> Translation into English: 'The ducks are flying again.'
> NOUN(nsubj) ADV VERB(root)

### Why?

- Extend multilingual analyses and benchmarks that require parsed data, like MultiBLiMP [5]
- Bootstrap UD treebank creation using existing data

### Next steps

- Evaluate annotation transfer and LLM methods on dependency parsed IGT data
- Does similarity to meta-language affect alignments?

[1] Georgi, R. A. (2016). *From Aari to Zulu: massively multilingual creation of language tools using interlinear glossed text* (Doctoral dissertation). University of Washington, Seattle, WA, USA.
[2] Ginn, M., Tjuatja, L., Rice, E., Marashian, A., Valentini, M., Xu, J., Neubig, G., & Palmer, A. (2026). *Massively Multilingual Joint Segmentation and Glossing* (arXiv:2601.10925). Preprint: arXiv.
[3] Ginn, M., Tjuatja, L., He, T., Rice, E., Neubig, G., Palmer, A., & Levin, L. (2024). GlossLM: A Massively Multilingual Corpus and Pretrained Model for Interlinear Glossed Text. In Y. Al-Onaizan, M. Bansal, & Y.N. Chen (Eds), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing* (pp. 12267–12286). Association for Computational Linguistics.
[4] Girrbach, L. (2023). Tü-CL at SIGMORPHON 2023: Straight-Through Gradient Estimation for Hard Attention. In G. Nicolai, E. Chodroff, F. Mailhot, & Ç. Çöltekin (Eds), *Proceedings of the 20th SIGMORPHON workshop on Computational Research in Phonetics, Phonology, and Morphology* (pp. 171–185). Association for Computational Linguistics.
[5] Jumelet, J., Weissweiler, L., Nivre, J., & Bisazza, A. (2026). MultiBLiMP 1.0: A Massively Multilingual Benchmark of Linguistic Minimal Pairs. *Transactions of the Association for Computational Linguistics*, 14, 193–216.

* Thank you to Anastasiia Panova for providing glossed texts in Gawarbati.